Ch 12 Final Project

November 23, 2020

1 Introduction

Chapter 12 discusses non-linear mechanics and chaos. While many systems obey deterministic equations of motion, not all systems have predictable future behaviors. Examples of systems that exhibit chaos whose equations of motion are non-linear include oscillating mechanical systems, the spread of disease, chemical reactions, fluid flow and lasers.

While some chaotic mechanical systems require new ways to describe their motion (like state-space orbits or bifurcation diagrams), others can be described in reasonable terms. An example of this would be the non-linear motion of a damped driven pendulum. A damped driven pendulum is apart of one of the two areas that describe chaos theory. One is the dissipative system, and the other is the nondissipative or "Hamilonian" system. While dissipative systems describe motion like the driven damped pendulum, nondissipative areas of chaos theory include astronomy and statistical mechanics problems.

First, to understand the problem I was given, we need to understand what it means for an equation to be linear or non-linear. For any system to exhibit chaos, its equation of motion must be non-linear. However, it is important to note, that just because a system is non-linear does not mean that it is guaranteed to exhibit chaos. Taylor, the author of "Classical Mechanics" textbook, starts by reviewing that "a differential equation is linear if it involves the dependent variable or variables and their derivatives only linearly" (458). We have seen many systems that are linear systems in this class. One example is the motion of a cart on a spring force. A non-linear example would be the equation for a pendulum, such as the example Taylor gives for a pendulum with an equation of motion (mass m, length L) is

$$mL^2\ddot{\phi} = -mgL\sin(\phi) \tag{1}$$

Since $\sin(\phi)$ is non-linear in ϕ, this is a non-linear equation for ϕ that is not extremely small. Though this system actually does not experience chaos. But, if we were to add a damping force $-bv = -bL\dot{\phi}$, this becomes the equation of motion for a driven damped pendulum:

$$mL^2\ddot{\phi} = -mgL\sin(\phi) - bL^2\dot{\phi} + LF(t) \tag{2}$$

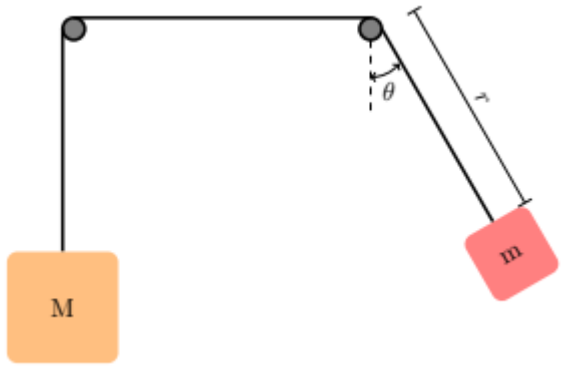

Figure 1: This is the swinging Atwood machine. One of the masses is free to oscillate over an angle θ. The length of the rope connected to mass m is given by the variable r.

Taylor explains that the parameters for a system to exhibit chaos is that its equation of motion is non-linear and "somewhat complicated" (460). Taylor also explains that what constitutes "somewhat complicated" goes beyond the scope of the book, but he does give us one last example of a chaotic system, which is the double pendulum. In the question that I was given, I have an Atwood machine where one of the masses is free to oscillate. This is essentially a pendulum which is attached to another mass through a pulley system. Through solving the different parts of my problem, I will be solving for the motion of $\ddot{r}$ and $\ddot{\theta}$, finding any equilibrium positions, and plotting the trajectory of the swinging mass over different time steps and varying parameters. My oscillating Atwood machine is seen here:

This system has a kinetic energy of

$$T = \frac{1}{2}M\dot{r}^2 + \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) \tag{3}$$

and a potential of

$$U = Mgr - mgr\cos(\theta) \tag{4}$$

2 Part A

In this part, we are asked to solve for the equations of motion of $\ddot{r}$ and $\ddot{\theta}$.

We can do this easily using the Lagrange equation $L = T - U$. This gives us

$$L = \frac{1}{2}M\dot{r}^2 + \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) - Mgr + mgr\cos(\theta) \tag{5}$$

Then we can utilize that

$$\frac{\partial L}{\partial r} - \frac{d}{dt}\frac{\partial L}{\partial \dot{r}} = 0 \tag{6}$$

Thus our first step is to take the partial derivative of L with respect to r. This gives us

$$\frac{\partial L}{\partial r} = mr\dot{\theta}^2 - Mg + mg\cos(\theta) \tag{7}$$

Then finding $\frac{d}{dt}\frac{\partial L}{\partial \dot{r}}$ we get

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{r}} = \ddot{r}(M+m) \tag{8}$$

Now we can plug these back into our equation to get that

$$0 = mr\dot{\theta}^2 - Mg + mg\cos(\theta) - \ddot{r}(M+m) \tag{9}$$

and rearranging to get $\ddot{r}$ alone we have

$$\ddot{r} = \frac{mr\dot{\theta}^2 - Mg + mg\cos(\theta)}{M+m} \tag{10}$$

The second part of this question is to find $\ddot{\theta}$. We can do this using the exact same process, except this time we have $\frac{\partial L}{\partial \theta} - \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = 0$. First, take the partial derivative of L with respect to θ gives

$$\frac{\partial L}{\partial \theta} = -mgr\sin(\theta) \tag{11}$$

Next, we take the partial derivate of L with respect to $\dot{\theta}$.

$$\frac{\partial L}{\partial \dot{\theta}} = mr^2\dot{\theta} \tag{12}$$

After this, we take the time derivative of the expression, remembering that both r and $\dot{\theta}$ have a time dependence and we get

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = 2m\dot{r}\dot{\theta} + mr^2\ddot{\theta} \tag{13}$$

Now we can finally plug this into our equation $\frac{\partial L}{\partial \theta} - \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = 0$ and rearrange for $\ddot{\theta}$ which gives us

$$0 = -mgr\sin(\theta) - 2m\dot{r}\dot{\theta} - mr^2\ddot{\theta} \tag{14}$$

and

$$\ddot{\theta} = -\frac{1}{r}(g\sin(\theta) + \frac{1}{r}2\dot{r}\dot{\theta}) \tag{15}$$

Thus we have our two equations of motion!

3 Part B

In this section, we are asked to simplify the equation for $\ddot{r}$ by letting $\mu = \frac{M}{m}$.

To solve this, let's remember our equation for $\ddot{r}$ and $\mu = \frac{M}{m}$, where $\ddot{r}$ is given by

$$\ddot{r} = \frac{mr\dot{\theta}^2 - Mg + mg\cos(\theta)}{M + m} \tag{16}$$

If we re-arrange our equation and let $M = \mu m$, then plugging this into our equation 16 becomes

$$\ddot{r} = \frac{mr\dot{\theta}^2 - \mu mg + mg\cos(\theta)}{\mu m + m} \tag{17}$$

where canceling all our variables m on the top and bottom gives us

$$\ddot{r} = \frac{r\dot{\theta}^2 - \mu g + g\cos(\theta)}{\mu + 1} \tag{18}$$

4 Part C

The system is decidedly non-linear, so you'll need to approach things in a non-linear fashion. Are there any equilibrium points, where the system would not move? Any particular requirements of the parameters for these points to exist?

To find the equilibrium points, we need to find where there is no acceleration or velocity. This means that we our values for $\ddot{r}$, $\dot{r}$, $\ddot{\theta}$, and $\dot{\theta}$ must all equal zero. First, I will look at our equation for $\ddot{\theta}$

$$\ddot{\theta} = -\frac{1}{r}(g\sin(\theta) + \frac{1}{r}2\dot{r}\dot{\theta}) \tag{19}$$

Letting $\ddot{\theta}$, $\dot{\theta}$, and $\dot{r}$ all equal zero we get

$$0 = -\frac{1}{r}(g\sin(\theta)) \tag{20}$$

For this to be true, we know $\frac{1}{r} = 0$ or $\sin(\theta) = 0$. This would mean that for our θ to be at an equilibrium point, thus our Atwood machine is not swinging, $\theta = 0$ or we are at point $r = 0$ where there is not any length to our rope.

Next, we also need to find where in our $\ddot{r}$ equation we have an equilibrium point. Let us remember that $\ddot{r}$ is given by

$$\ddot{r} = \frac{r\dot{\theta}^2 - \mu g + g\cos(\theta)}{\mu + 1} \tag{21}$$

Setting $\ddot{r}$ and $\dot{\theta}$ equal to zero we end up with

$$0 = \frac{-\mu g + g\cos(\theta)}{\mu + 1} \tag{22}$$

Now, we can apply the fact that we know for our system to be at an equilibrium position $\theta = 0$. Thus $\cos(\theta) = 1$ and pulling out gravity our equation becomes

$$0 = \frac{-\mu + 1}{\mu + 1} \tag{23}$$

To solve this equation, we can multiply both sides by $\mu + 1$ and get

$$0 = (\mu + 1)(-\mu + 1) \tag{24}$$

Thus through a little bit of algebra, we know our system will be at equilibrium when $\mu = 1$. Remembering that $\mu = \frac{M}{m}$, this means our system will only have an equilibrium point if the two masses of our system are equal ($M = m$).

5 Part D

When $\mu = 7.7$:

i. To get an idea for the behavior of the system, you'll need to solve things numerically. Taking $\mu = 7.7$, plot the trajectory of the swinging mass over 40 seconds if it starts with $r_0 = 1$, $\theta_0 = \frac{\pi}{2}$, $r = 0$, $\theta = 0$.

ii. To check for signs of chaos, plot another trajectory whose initial conditions are identical except for $r_0 = 1.005$. Determine the distance between your two objects at each point in time and use this to recreate a plot similar to Figure 12.13, with the log of the distance between the points on the vertical axis and time on the horizontal. This would be the plot one would use to determine the Liapanov exponent. Does your plot support the idea that this motion is chaotic?

All of this we can do with the help of python. To do this, I created a 'while' loop, and had the loop run for $t < 41$. In the loop, I had the equations update for $\ddot{r}$, $\ddot{\theta}$, $\dot{r}$, $\dot{\theta}$, r, θ and t. I then had all of my r, θ, and time values append to a list. Finally, I created a polar plot to show the motion of the system which looks like It is important to note that the direction down towards the ground is actually to the right in this figure. This is because the default for $\theta = 0$ is to the right for python.

Next, to check for chaos, I plotted another graph with the same initial conditions except for $r_0 = 1.005$. This is shown by Then taking the difference between the two position of the two plots and plotting over time to compare to Figure 12 in the book, we get Here, we see that the plot appears to be a sine graph and then tapper off to zero. We can conclude here that the drive frequency ω is changing, just like Figure 12 in Taylor's book. This means that as the initial condition changes, the non-linearity of the motion becomes important. If the system is not chaotic, then we should have a graph that approaches and settles into a sinusoidal oscillations. We should also see that as time increases the position $\theta(t)$ should agree with each other. Meaning that if a system does not experience chaos, the positions for θ should start to equal each other after long time periods. In this problem, this did not happen and the difference between the two values did not converge. Thus we can conclude that this system experiences chaos.

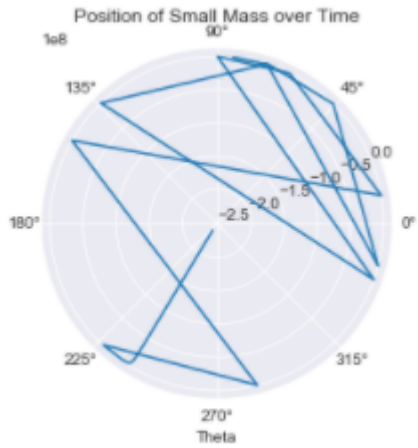

Figure 2: This is a polar plot of the swinging Atwood machine. Here, the direction down towards the ground is actually to the right, because the default $\theta = 0$ is to the right.

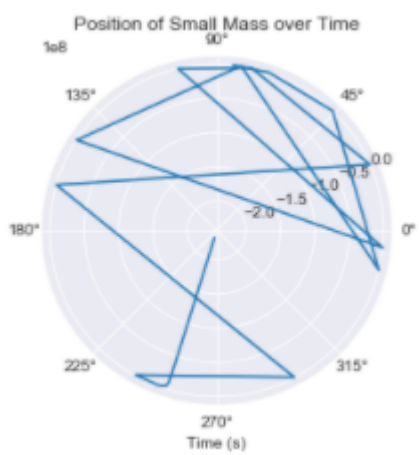

Figure 3: This is a polar plot of the swinging Atwood machine with $r_0 = 1.005$. Here, the direction down towards the ground is actually to the right, because the default $\theta = 0$ is to the right.

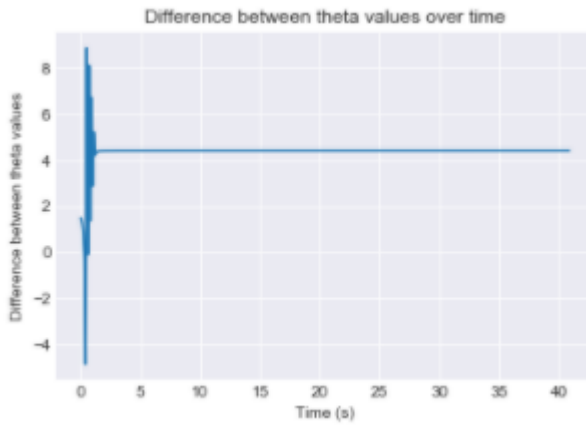

Figure 4: This is the difference between the theta values in Figure 1 and 2 over time. It appears to be sinusoidal and then eventually go to zero.

6 Part e

For $\mu = 3$, compute all the parts for part d for $\mu = 3$ and a run time of 120 second.

To start, using the exact same process as in part d, I can change my values of μ and t to get Figure 5: Then using a $r = 1.005$, I found the plot to be (Figure 6):

These graphs did not appear to look quite as chaotic as my first two graphs. To test this hypothesis, I moved on to the second part of this problem and calculated the difference between the two positions. This was easily accomplished in python. I noticed that the positions of the two graphs here also did not approach the same number, meaning that they exhibit a chaotic system. Then, graphing the solution looks like From this graph, we see that we have what looks like a sine function that is gradually changing. Much like in part d, this is showing a chaotic system. A system that is not chaotic should eventually approach a normal non-changing sine graph.

7 Part f

Repeat all steps done in part d, this time with $\mu = 10.63$ over a time of 120 seconds.

Here, we can follow the exact same process as part d and e. All we have to do is use the code that we already have and change the values of $\mu = 10.63$ and run the loop over 120 seconds. For an initial $r = 1$, this will give us a plot that looks like (Figure 7): Then changing the initial conditions slightly so that $r = 1.005$, we get a polar plot that looks like

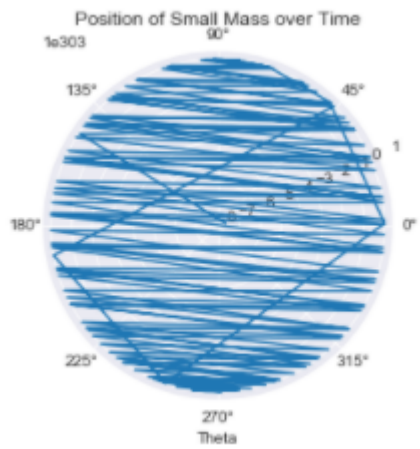

Figure 5: Using a time of 120 seconds and $r = 1$, this is the polar plot of the system with a $\mu = 3$.

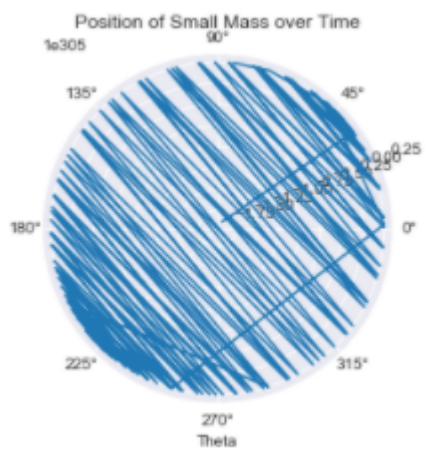

Figure 6: Using a time of 120 seconds and $r = 1.005$, this is the polar plot of the system with a $\mu = 3$.

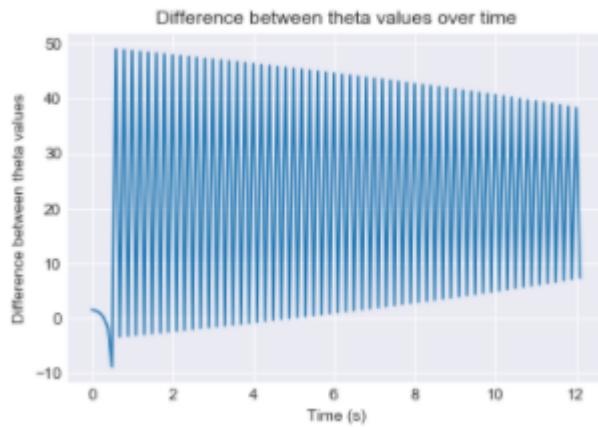

Figure 7: This is the difference of the position of Figure 5 and 6 through time. It shows a choatic system with a dampening sine graph.

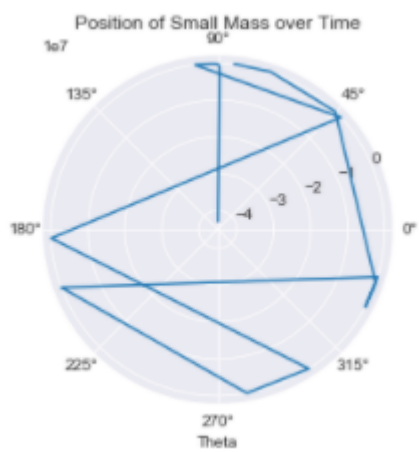

Figure 8: Using a time of 120 seconds and $r = 1.0$, this is the polar plot of the system with a $\mu = 10.63$

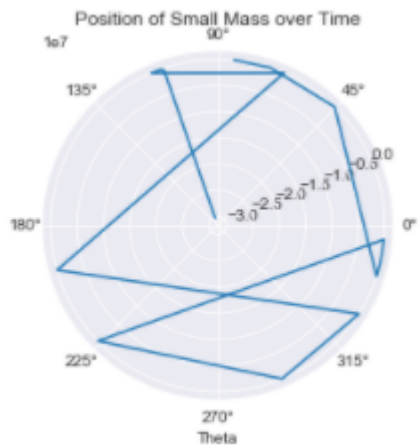

Figure 9: Using a time of 120 seconds and $r = 1.005$, this is the polar plot of the system with a $\mu = 10.63$

Finally, when we compute the distance between the two, we see that the first situation approaches a value of about 1.56028 and the second situation approaches 1.90442. Already, we can tell that this system is also chaotic. However, plotting the difference, we get the graph

Finally, we can conclude from the graph that our system does not approach a sinusoidal function over a long time period and is therefore chaotic.

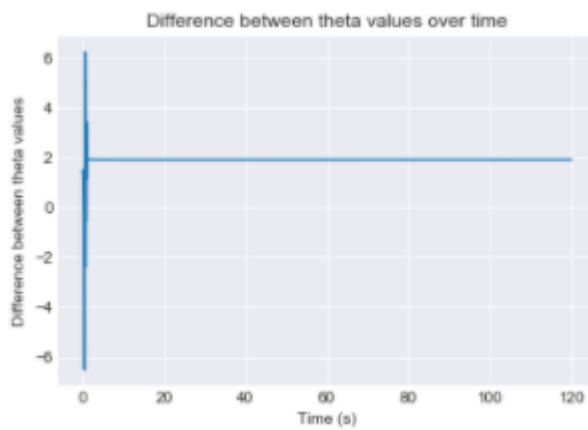

Figure 10: Here we see the difference in position of our system for a starting $r_0 = 1$ and $r_0 = 1.005$ for a given $\mu = 10.63$ over 120 seconds. Here, we see a chaotic system that does not go to a sinusoidal function.